

A study about the implied meanings of responses
Each of the responses in the following sentences means yes or no, choose the right one.

What is your unique Prolific ID? *

Which email can we send your Completion Code to once you finish this form? *

Esther asked "Is the dress perfect?" and Juan responded "it fits like a glove.", which means

☐ Yes

☐ No

Esther asked "Aren't you scared?" and Juan responded "Sometimes.", which means

☐ Yes

☐ No

Esther asked "Are you focused?" and Juan responded "Yes.", which means *

☐ Yes

☐ No

Esther asked "Is it a difficult test?" and Juan responded "It would be a piece of cake", which means

☐ Yes

☐ No

(a) The start of the Google form participants are asked to fill out for the human study.

(b) Part of the Google form the participants are asked to fill out. The second question in this image is part of the attention test. Juan's response does not contain an implicature but simply gives away the correct answer.

Figure 6: Screenshots of the Google form participants fill out as part of the implicature study.

Table 10: The performance of the human annotators on the subsets of the test set. Subset 1 through 4 are non-overlapping and cover the whole test set. Annotator X for subset Y might be a different human than annotator X for subset Z. IAA is the average pairwise inter-annotator agreement (Cohen's kappa coefficient) between annotators per subset.

Annotator	1	2	3	4	5	Mean	Best	Worst	IAA
Subset 1	86.0%	92.0%	90.7%	90.6%	86.0%	89.1%	92.0%	86.0%	0.73
Subset 2	84.7%	83.3%	87.3%	86.0%	86.0%	85.5%	87.3%	83.3%	0.64
Subset 3	84.0%	85.3%	88.0%	86.0%	82.7%	85.2%	88.0%	82.7%	0.78
Subset 4	85.3%	82.7%	84.0%	82.0%	92.0%	85.2%	92.0%	82.0%	0.71
Total	-	-	-	-	-	86.2%	89.8%	83.5%	0.72
Std	-	-	-	-	-	2.3	2.2	1.5	0.1

Human source of disagreement with ground-truth. We do an analysis of the source of disagreement with the ground-truth. We explicitly do not call this error, as in some cases examples might allow multiple interpretations, and both could be right. In other cases, the ground-truth might be wrong.

Annotation errors and multiple interpretations: We analyse the examples for which most humans choose a different answer than the ground-truth. For 30 out of 600 examples in the test set, only one or zero people choose the same answer as the ground-truth. Of these examples, most are annotated wrongly (18 of 30). For example: 'Are you busy?', 'I'm drowning in work.', implicature: 'no'. Some are examples that can have multiple different interpretations (12 of 18), and the ground-truth answer likely just chooses one that is unnatural to humans. For example: 'You don't remember them?', 'Leave me alone!', implicature: 'yes'. 6 of the 30 examples are particularised, and 1 is generalised.

Examples for which all humans agree with the ground-truth: There are 409 out of 600 examples that all humans get correct. This set of examples contains most of the generalised implicatures (39 out of 47). These contain 58 out of 94 particularised examples.

Examples most humans agree with the ground-truth: When we look at examples that 3 or more humans got correct, that comprises most of the test set (530 of 600), and all of the generalised examples (47 of 47). This subset has 78 of 94 particularised examples, so for 16 particularised examples 3 or more humans disagree with the ground-truth.

I Comparison with BIG-bench implicatures task

One of the BIG-bench tasks is related to the task in this work⁹. It uses the same underlying dataset we use (George and Mamidi, 2020). With the below we aim to discuss our contribution in light of the BIG-bench result. To summarise; the methodology used by the BIG-bench task contributors has limitations, which call into question the validity of their claims. Further, some of the BIG-bench results are irreproducible due to missing details in the tech report and the use of proprietary models. Considering this, our work is an important contribution validating the BIG-bench results in a reproducible and methodologically sound way, and above that providing insight into what aspects of LLM training are crucial for the ability to do pragmatic inferences.

Limitations of the methodological approach of the task contributors in BIG-bench implicatures.

Our benchmark has 30% more data, which the BIG-bench task contributors discard. In this section we motivate the crucial importance of that data for evaluating implicature understanding (Section I.1), and why BIG-bench in turn might be overestimating the performance of LLMs on implicature resolution (Section I.2). Moreover, the human performance on the BIG-bench task indicates low quality human annotation, which we will also elaborate upon below, noting that this is impossible to verify because the BIG-bench report does not detail how the evaluation was done for this task (Section I.3).

I.1 Discarding ambiguous examples

The BIG-bench task preprocesses the 1001 examples that George and Mamidi (2020) curate by keeping only yes/no questions, discarding any examples that are ambiguous according to the task contributors, and discarding remaining examples to create a 50/50 distribution of yes/no answers. This leaves them with 492 examples. Of these examples, 81 appear in our development set and the remaining 411 appear in our test set. Our test set has 600 examples, so BIG-bench effectively discarded 189 ambiguous examples compared to our test set; a bit more than 30% of the benchmark. To illustrate the importance of not discarding this data, we cherry picked a few examples that the BIG-bench authors discarded from the data.

- Utterance: "Can you lend me hundred dollars?", Response: "Is this supposed to be some kind of a joke?", Implicature: "No"
- Utterance: "Do you know, how long is Uncle Arthur staying with us?", Response: "Ask your father.", Implicature: "No"

Indeed, these examples are ambiguous. Asking whether the request for a hundred dollars is a joke does not literally mean you're saying no to the request. The response "ask your father" does not mean the speaker does not actually know, maybe they just do not want to respond. The humans in our study all infer the intended ground truth implicature. This shows a general property of implicatures; they are ambiguous, but often humans do infer the intended meaning. Ambiguity is not a discrete property. Some examples may be so vague that no one gets it. The following are examples the BIG-bench task discards that the humans in our study did struggle with:

- Utterance: "Got any more of those?", Response: "Nothing I'm at liberty to reveal here.", Implicature: "Yes"
- Utterance: "Have you finished sight-seeing?", Response: "Sorry. I should've come to see you first.", Implicature: "Yes"

In the first of these the implicature is "yes" because the person responding is implying that they do have more, they just cannot reveal them. Otherwise they would most likely simply say no. In the second example it feels more natural that someone says this when they are finished sight-seeing, otherwise they would've probably said something to the effect of "I'm still out, but I'm sorry.". In any case, humans in our study did not understand these responses like that. This illustrates another aspect of implicature; sometimes communication will go wrong over it. Removing implicatures that are ambiguous though, defeats the purpose of the task, as they are all ambiguous to a certain degree. The purpose of this study is to compare how humans resolve this type of non-literal language

⁹https://github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/implicatures

compared to how models do it. The human baseline of 86% accuracy that humans achieve on our test set deals more naturally with examples that are too ambiguous for models to understand than discarding examples based on the subjective opinion of a few people.

I.2 Overestimation of performance on implicature understanding

On the overlapping part of our test set and theirs the humans in our study achieve 92.8% accuracy. The best model on the BIG-bench task is PaLM, achieving a zero-shot performance of 64.4%. Note that this performance is on their full test set (not the overlapping part) and hence not directly comparable. Nonetheless, the missing examples are randomly sampled for our development set, and we can be pretty confident this number indicates a large gap with human performance. Two-shot PaLM comes very close to human performance with 91.7% accuracy, but of course this does not take into account the 189 more challenging examples that are part of our benchmark. Humans achieve 71.9% performance on this subset of ambiguous data, indicating that these data are more difficult than average, but nonetheless performance is higher than random. Without access to the models used to evaluate the BIG-bench task we cannot say anything for certain, but we expect the performance of PaLM to be overestimated as it does not account for a large part of the type of implicatures found in natural discourse; ambiguous ones.

I.3 Other limitations

Poor quality human evaluation. The average human evaluator on BIG-bench implicatures achieves around 82% performance (where ours achieves on average 86% on a more challenging dataset), and their human best rater achieves 100% (where our human best is 92%). This difference between human average and best hints at poor quality average rating. This is impossible to verify because there is no information in the BIG-bench tech report on how the human evaluation was done exactly, or even which examples were evaluated.

No fine-tuned models. BIG-bench uses only base LLMs and no SotA fine-tuning methods. A question that remains is therefore, what aspects of LLMs contribute to their performance on implicatures? In our work we find that implicature performance emerges at a much smaller scale in models instruction fine-tuned at the example level, and that scale and prompting techniques are important.